

Capture-the-Flag

The lexicon, tiered

Two questions sit under this project, and the index line on the branch page asks both at once.

“what makes a CTF challenge difficult? how can we ensure the effectiveness of competitive cybersecurity training in the era of agent solvers?”

website, Problem-Solving, the branch index

The first question is the one the six phases answer. The second is the one that decides whether the answer still matters in three years. The position the project takes on it is short, and it is stated on the poster.

“The hard part of a CTF is the thinking, not the typing. AI agents can now run the tools, so what is worth teaching and testing is the set of mental moves a solver makes.”

poster draft, SIGCSE Virtual 2026, headline claim

Everything below is the vocabulary that position needs. Active entries carry all four fields, because they are the ones I have to define on the spot to a designer, to a student, or to a reviewer who has decided in advance that this is a tooling paper. Working entries carry a definition and the discrimination. Peripheral entries carry a definition and a pointer to where the term belongs.

Six terms are used in two senses across my own files, and difficulty is used in three. Each sits inside the entry it belongs to, in a marked box, because a term whose uses disagree is the entry most worth having.

Contents

1	Active	3
1.1	The solve	3
1.2	The six phases	5
1.3	Getting the data	7
1.4	The design side	9
1.5	The agent in the room	11
2	Working	12
3	Peripheral	14
4	Where two locators disagree	15
5	What crosses into the shared lexicon	15

1 Active

1.1 The solve

solve

“A solver cycles through them, and the cycling is the solve.”

website, Capture-the-Flag, two deliberate choices

Definition: One participant’s whole attempt at one challenge, from opening it to submitting a flag or abandoning it, treated as the unit of analysis.

Distinguishes it from: A session, which is whatever the recording happens to cover, and a competition, which is many solves by many people at once. The solve is bounded by the challenge, not by the clock or by the file.

Operational test: Does the stretch of behaviour have one challenge and one person at both ends of it? If either changes partway, that is two solves.

In my project: One recorded solve from YouTube, annotated into a graph of questions and operations. That single artifact is what carries H1 on the poster, so it is a solve in the strict sense and not a highlight reel.

mental operation

“A mental operation is a step that takes the solver from one or more questions they were holding to one or more new questions.”

Prep notes, sections/qa_log.tex, Q1

“a number saying how sure we are that the step really happened, and a note of who made it”

same

Definition: A typed, directed step from a set of held questions to a set of new ones, carrying a confidence and an attribution to the person or to the tool that made it.

Distinguishes it from: A tool invocation. A command is an act in the world; an operation is the step in the head that decided to run it. The two are recorded against each other on purpose, since the attribution field is what lets an agent-assisted solve be separated from an unassisted one.

Operational test: Can you name the questions it consumed and the questions it produced? If the step cannot be written as questions in and questions out, it is an action and it belongs in the trace, not in the operation set.

In my project: The hyperedge of the framework document, and the label the annotators apply. The reason it has an arity above one is that a solver often resolves two questions with one step, which an ordinary directed graph cannot represent.

An operation is a definition problem before it is a measurement problem

The framework treats the missing definition as the work itself: “There is no commonly accepted definition of a mental operation, and that gap is not a detail to defer. The strength of any

hypothesis stated in terms of mental operations depends on how the term is fixed”, `core idea/main.tex`, on converging on an account of mental operation.

The poster treats it as settled enough to build on: “When a person solves a CTF, they use identifiable mental operations”, poster draft, H1.

Both are mine and both are current. The poster is an empirical claim about whether operations can be recovered from a solve; the framework is a claim about whether the construct is fixed enough for that recovery to mean anything. A reviewer who reads them together will ask which one the study rests on, so the answer is written here rather than found in the room: H1 is a claim about inter-rater recoverability under a fixed grain, not a claim that the construct has been settled.

question

“whose nodes are questions and whose hyperedges are mental operations, reconstructed from a solver’s action trace”

`core idea/main.tex`, abstract

Definition: The node of the solve graph. What the solver is trying to determine at a given moment, stated as a question rather than as a state.

Distinguishes it from: A subgoal. A subgoal is something to achieve and is satisfied or not; a question is something to determine and can be answered wrongly, which is what makes the backtracking in a real solve representable.

Operational test: Can it be written with a question mark and answered by evidence the solver could obtain? If it can only be written as an imperative, it is a step, not a node.

In my project: The annotation protocol asks the annotator to name the question before naming the operation, so that operations are recovered as the links between questions rather than as a free-standing list of verbs.

the four relation types

“one where the new question is built out of the old ones as parts of a whole; one where answering the old questions is what lets you reach the new one; one where the questions simply come up together, with no claim that one caused the other or came first; and one where the new question cannot be answered until the old ones are answered first, which gives the order in which things have to be done.”

Prep notes, `sections/qa_log.tex`, Q1

Definition: The label on an operation. Compositional, causal, co-occurrence, dependency. The dependency edges alone form the acyclic prerequisite structure of the solve.

Distinguishes it from: Time order. Two questions that arise together carry a co-occurrence label and no claim of sequence, which is why the annotation stays usable on a trace where the timestamps are coarse.

Operational test: Ask whether the earlier question had to be answered first. Yes gives dependency; if answering it also explains why the later one is answerable, causal; if the later question is assembled out of the earlier ones, compositional; if neither, co-occurrence.

In my project: Criticality is defined on the dependency graph alone, as the value of information

of a question, which is why a question can be a prerequisite without being a cause.

1.2 The six phases

phase

“Orientation. Information gathering and reconnaissance. Hypothesis formation. Strategy selection and execution. Monitoring and regulation. Verification and transfer.”

website, Capture-the-Flag, the six phases

Definition: A band of a solve, broad enough that two annotators agree on where it starts. Operations live inside phases; the phase is the coding unit and the operation is what the phase contains.

Distinguishes it from: A stage in a scripted attack chain. A phase is a description of the solver, not of the challenge, and phases two to four recur many times in one solve while a chain stage does not.

Operational test: Would two coders, given only the recording and the phase list, mark the same boundary within a tolerance you fixed in advance? If not, the phase set is too fine.

In my project: The taxonomy of record numbers them zero through five with a cross-cutting layer beneath. The page gives the six names; the contents are below.

Phase	What it contains
0. Orientation	Problem framing (crypto, forensics, web); scope assessment; resource inventory, what do I know and what tools or documentation exist; belief activation, skip it against worth investing time
1. Information gathering and reconnaissance	Reading and comprehending provided materials; environmental exploration, scanning, traversal, inspecting artifacts; pattern recognition, this looks like base64; signal and noise discrimination
2. Hypothesis formation	Abductive reasoning; analogy to prior experience; decomposition into sub-problems; mental simulation, if I try X what happens
3. Strategy selection and execution	Tool selection; command formulation; parameter tuning; iterative refinement, try, observe, adjust
4. Monitoring and regulation	Progress assessment; strategy switching; time management; help-seeking decisions; frustration management
5. Verification and transfer	Solution validation, flag-format checks; error detection and diagnosis; knowledge consolidation for the next challenge
Cross-cutting: beliefs and affect	Confidence calibration; competition-driven rushing; intimidation by unfamiliar domains; the satisfaction and reward experience

the loop

“Gathering, hypothesising and executing don’t run once in order.”

website, Capture-the-Flag, two deliberate choices

Definition: Phases one through three, taken as a cycle rather than a sequence. The cycling is the object of study, not an irregularity in it.

Distinguishes it from: Pólya’s four phases, which run once in order. That model is the ancestor of this one and its strict sequence is the property this one drops.

Operational test: Count the transitions from phase three back to phase one in a coded solve. If the count is zero across a corpus, the phases are being coded as a script.

In my project: The loop is a post-hoc amendment to the taxonomy, agreed after the first coding pass, and the framework copy rather than the original writeup is the version that carries it.

belief systems and affective states

“What the solver thinks this kind of challenge usually requires, and how they feel about being stuck, colour every phase above.”

website, Capture-the-Flag, the six phases

Definition: The cross-cutting layer. What the solver expects this kind of challenge to demand, and the affect that accompanies being stuck in it. Coded alongside the phases, never as one of them.

Distinguishes it from: Phase four. Monitoring and regulation is something the solver does; a belief is something the solver holds, and it changes the cost of every operation above without appearing as a step anywhere.

Operational test: Could the item be true of the solver before the challenge was opened? If yes, it is a belief, not a phase.

In my project: Schoenfeld’s fourth knowledge category is the ancestor of this layer, and it is what explains the instruction-skipping pattern the designers report: a belief failure about what a competition rewards, compounded by a control failure.

grain

“Finer metacognitive coding schemes have a history of collapsing under inter-rater disagreement, so the phases stay broad and the specific operations live inside them. That’s a reliability decision, not an omission.”

website, Capture-the-Flag, two deliberate choices

Definition: The size of the unit an operation is coded at. There is no principled unit, so the grain is chosen per purpose and held fixed for the study that uses it.

Distinguishes it from: Resolution in the ordinary sense. A finer grain buys detail and spends agreement, and past attempts have spent more than they bought, which is why the choice is made once and written down rather than tuned toward a better-looking result.

Operational test: Is the grain stated in the protocol before the first solve is coded? A grain

chosen after seeing the disagreement figures is a selection, not a decision.

In my project: The annotation protocol fixes it at the smallest unit of action, which is the deliberate fix the operations essay argues for. The reliability cost of not fixing it was the first thing raised against the protocol in review, which is exactly what the essay predicted.

control

“Knowing an operation is not using it well.”

website, Mental World Modeling, Next

Definition: The layer that decides which operation to run next, notices that a line is not working, and decides when to abandon it. Coded separately from the operations it governs.

Distinguishes it from: The operation set. An operation is what is deployed; control is when it is deployed and when it stops. Schoenfeld’s result is that control, not the stock of heuristics, is what separates experts from novices, so a taxonomy that folds control into the operation list has lost the discriminating half.

Operational test: Can the item be performed without reference to how the solve is going? If it can, it is an operation. If it needs the state of the attempt, it is control.

In my project: Tool selection and knowing when to stop are the two concrete versions in this setting, and both look more like optimal stopping sitting on top of the solve than like steps inside it.

Tool selection sits in two places

The taxonomy puts it inside a phase: tool selection is the first item under Phase 3, strategy selection and execution, `core_idea/sections/part1_operations.tex`.

The cluster’s open-problems page puts it above the phases: tool selection and knowing when to stop “look less like steps inside a solve and more like an optimal-stopping problem sitting on top of it”, website, Mental World Modeling, Next.

Both are defensible and they produce different coding schemes. Coded inside phase three, choosing a tool is one operation among four. Coded as control, it is the thing the study is actually about, since it is the operation an agent solver does not make the same way a person does. The taxonomy of record currently keeps it in phase three and lists it as a first-class operation, which is the compromise, not the resolution.

1.3 Getting the data

elicitation

Definition: Any method for recovering what a solver was doing mentally from something observable. Six are in play and five are ruled out for this setting, each for its own reason.

Distinguishes it from: Instrumentation. Logging what happened is cheap and is not elicitation; elicitation is what recovers the why, and every method that does it costs either intrusion or invention.

Operational test: Does the method require the solver to say something? If not, it is instrumentation and it will not answer a question about reasoning.

In my project: The workable combination is one method plus one instrument, stimulated recall anchored to submission logs, and the reason is that the competition setting rules out the gold standard.

Method	What it gives	The verdict here
Concurrent think-aloud	Real-time verbalisation during the solve; the gold standard for access to process	Too intrusive for a timed competition; usable in a lab, outside the event
Retrospective think-aloud	Narration after the fact, no interference	Invites invention; the solver reconstructs rather than reports
Experience sampling	Self-report at trigger points, such as a short popup after a submission	Interrupts; and items get skipped under time pressure, so the missing data is biased toward the most competitive solvers
Cognitive task analysis interview	Structured recovery of one concrete incident, in deepening passes	Expensive, and depends on having an expert on hand
Platform logs	Timestamped submissions, selection order, error patterns, at large scale	Cheap, and tells you behaviour only
Stimulated recall	Narration cued by the solver's own screen replay	The workable one, once it is anchored to the logs

stimulated recall

“replay the solver their own session as a retrieval cue”

website, Capture-the-Flag, getting the data

“anchored to submission logs, so the recall has timestamps to hang on”

same

Definition: Retrospective narration in which the solver's own recording is played back as the retrieval cue, with log events used as the anchors the questions are asked against.

Distinguishes it from: Retrospective think-aloud, which is the same timing with no cue. The cue is the whole difference: without it the solver reconstructs a plausible account, with it the solver is answering about a specific moment that is on the screen in front of them.

Operational test: Is there a cue on the screen at the moment the question is asked? If the interviewer is asking from memory, it is retrospective think-aloud however it was labelled.

In my project: The recall interval is minimised, same session and ideally immediately after, and probes are anchored on log events. The worked example of a probe is a run of three wrong submissions, asked about as a run rather than as three separate events.

One method or two

The website page lists them as separate items with separate verdicts: “Retrospective think-aloud invites invention”, and then, two sentences later, “The workable combination is stimulated recall”, website, Capture-the-Flag.

The methods writeup pairs them under one heading, retrospective think-aloud and stimulated recall together, and gives the pair one verdict, strong, `core idea/sections/part1_methods.tex`. The difference is not cosmetic. If they are one method, the invention risk the page assigns to retrospective think-aloud applies to the method the study actually uses, and the protocol has to say what controls it. If they are two, the cue is the control, and that has to be stated as the reason. The page’s split is the one to keep, and the cue is the reason.

submission log

“Platform logs are cheap but only tell you behaviour.”

website, Capture-the-Flag, getting the data

Definition: The competition platform’s timestamped record of flag submissions, correct and incorrect, per participant per challenge. Used here as the anchor track for recall rather than as evidence in its own right.

Distinguishes it from: A screen recording, which is continuous and carries no events. The log is what turns the recording into something questions can be addressed to, since it supplies the moments worth asking about.

Operational test: Does the record have discrete timestamped events tied to a participant and a challenge? If it is a continuous stream, it is a recording, and it needs the log to be interrogable.

In my project: CTFd is the platform the anchors come from. At scale the same records support inference about strategy from selection order and inter-submission timing without any self-report, which is what the picoCTF submission studies do at half a million submissions; that is the complement to recall, not a substitute for it.

1.4 The design side

designer

“designers hold rich tacit knowledge about what makes challenges hard but lack explicit vocabulary for the cognitive operations they test”

`core idea/sections/part2_design.tex`, connecting operations to design intent

Definition: The person who builds the challenge. The second population of the study, interviewed about one concrete challenge of their own rather than about design in general.

Distinguishes it from: An instructor. A designer sets the challenge and never sees most of the solvers; the prediction task is what recovers whether they modelled those solvers correctly anyway.

Operational test: Did they choose what the challenge would demand? If they only assigned it, they are an instructor and their predictions are about students, not about the artifact.

In my project: Five pilot interviews, and the half of the hypothesis that ties cognition to design. It is why the pilots interview designers and not only solvers.

predict-then-test

“a designer who models solver operations should be able to predict solver behavior”

Prep notes, sections/qa_log.tex, Q2

Definition: The scoring paradigm for the designer half. The designer predicts where solvers will stall and what they will try; those predictions are scored against what the solvers did.

Distinguishes it from: Asking a designer to rate their own challenge’s difficulty. A rating is a claim about the artifact; a prediction is a claim about people, and only the second one can be wrong in a way that is informative.

Operational test: Is the prediction made before the solver data exists, and recorded? If it is elicited afterward it is a rationalisation.

In my project: Borrowed from the teacher judgment-accuracy literature, and run on the designer’s own challenge rather than on a challenge set, so that a wrong prediction cannot be attributed to unfamiliarity.

bottleneck

Definition: A point in a challenge that only lets through solvers who perform a particular operation, with the hint system placed to rescue the rest. A design lever for making one operation necessary rather than optional.

Distinguishes it from: Difficulty in general. A bottleneck is targeted at an operation; raising the technical demand of every step raises difficulty without selecting for anything.

Operational test: Name the operation the gate requires. If it cannot be named, the step is hard rather than selective.

In my project: The strongest pilot transcript describes a branching structure narrowing from five candidate paths to one, gated on noticing a planted anomaly, with hints existing to rescue the solvers who do not notice. That is a designer targeting an operation without a word for it, which is the whole case for supplying the vocabulary.

hint ladder

Definition: An ordered set of hints of increasing information content, with the placement of the first tier calibrated against the point where a novice abandons the challenge.

Distinguishes it from: A single hint, which either arrives too early and removes the operation, or too late and loses the solver. The ladder is what makes the amount of assistance a design variable rather than a binary.

Operational test: Does each tier collapse a different part of the search space? If two tiers reveal the same thing at different wordings, it is one hint written twice.

In my project: One designer calibrates the ladder by solving his own challenge as if he were a beginner, having observed that students abandon a brute force attempt within five to ten minutes. His tiers escalate from the encoding, to the cipher stack on top of it, to the wordlist theme that collapses the search space. That is playtesting with a sample of one, and it is the cheap end of

an established method.

difficulty

“The difficulty of a problem, for a solver θ , is a vector functional $D_\theta(P)$ rather than a single number, because its three sources are not commensurable.”

core idea/main.tex, Definitions

Definition: A vector over three sources: solution size, the length of the optimal solution; concept load, the number of concepts held and related at once; and search burden, the effort of finding the operations rather than applying them.

Distinguishes it from: A rating. A rating is one number and hides which of the three sources moved, which is precisely the information a designer needs in order to change one of them.

Operational test: Change one source and hold the other two. If the resulting challenge is described as equally difficult, the vector is not being used.

In my project: Difficulty and ability are two sides of one relation, so the vector is indexed by the solver. Where no solution exists, difficulty is maximal rather than undefined, because certifying that a closed problem has no solution is itself a co-NP task.

Difficulty in three senses, all of them mine

Structural: the vector over solution size, concept load and search burden, core idea/main.tex, Definitions.

Relational: a designer’s account in which the same artifact is trivial or impossible depending on what the student notices, so his levers are hint placement and what is left observable, pilot interview 02-02.

Experiential: hills and valleys, difficulty modulated across a sequence of challenges rather than within one task, pilot interview 02-03.

The three do not contradict each other, but they do not compose either. The vector is a property of a problem and a solver; the relational account is a property of a problem and what was left visible; the experiential account is a property of an ordered set of problems. A question about difficulty is underspecified until it says which of the three is meant, and the interview guide asks all three without saying so.

1.5 The agent in the room

agent solver

“AI attacker agents can now solve many CTF challenges on their own. This lowers the skill floor: work that used to take a trained person can be done by running a tool. Banning the tools is a losing game; the better response is to adapt CTF to a world where the tools exist.”

poster draft, SIGCSE Virtual 2026, introduction

Definition: An autonomous system that completes a challenge end to end without a person choosing the steps. Treated here as a condition of the environment, not as a subject of the study.

Distinguishes it from: An assistive tool used during a human solve, which appears in the attribution field of an operation and leaves the solve a human solve. The agent solver replaces

the solve; the assistant is inside it.

Operational test: Who chose the next step? If no person did, at any point, it is an agent solve.

In my project: The reason the study exists rather than a thing the study measures. The claim is that the operations worth teaching are the ones an agent does not make the same way, and that claim is what the designer-side question about human against agent divergence is written to test.

human against agent divergence

“What is hard for a person is not what is hard for an agent.”

Prep notes, sections/poster_draft.tex, early results

Definition: The gap between the operations a challenge demands of a person and the operations it demands of an agent. Named by a designer before it was named by us.

Distinguishes it from: A capability gap. This is not about which of the two is better; it is about which operations each finds expensive, and those sets are different in both directions.

Operational test: Take one challenge and ask which operations each would handle well and which poorly. If the two answers are the same list, the challenge is not discriminating.

In my project: The one participant-voiced statement of it across the five pilots: humans in a competition skim and rush by choice, while an agent would dissect and digest the entire instructions. His conclusion is that instructions as a source of difficulty penalise humans differentially, which is a design finding, not a complaint.

self-management as difficulty

“Do any of your challenges derive difficulty from self-management rather than technical content”

core idea/sections/part2_design.tex, designer question bank, E

Definition: Difficulty that comes from regulating the attempt rather than from the technical content: deciding when to abandon a line, resisting the pull of a sunk cost, tolerating being stuck.

Distinguishes it from: Technical difficulty, which an agent absorbs. This is the residue the poster’s claim depends on, so it is the category that has to be measurable if the claim is to be worth anything.

Operational test: Would the challenge get easier for a solver who had unlimited patience and no time pressure, holding knowledge constant? If yes, part of its difficulty is self-management.

In my project: One of the five items on the beliefs and affect scale asks it directly from the student side: I kept working longer than I should have because I had already invested time.

2 Working

inter-rater agreement

Definition: The proportion of coded units on which independent coders assign the same label, reported alongside a chance-corrected statistic. The criterion the whole taxonomy is defended by.

Distinguishes it from: Accuracy. There is no ground truth here, so agreement is the only

available evidence that the construct is a construct; a scheme nobody can code reliably is not one.

coding scheme

Definition: The fixed list of labels, their definitions, and the boundary rules an annotator applies to a recording. Defended by reliability and by predictive use, not by claimed naturalness.

Distinguishes it from: The taxonomy, which is the set of categories. The scheme is the taxonomy plus everything needed to apply it to a case, which is where the disagreement actually comes from.

annotation protocol

Definition: The written procedure the annotators follow: the grain, the order of naming, the vocabulary sources, and the agreement checks. The place the grain decision is recorded.

Distinguishes it from: The instrument. Astrolabe is what the protocol runs in; the protocol is what makes two runs of it comparable.

Astrolabe

Definition: The local annotation application this work runs in. It takes transcripts, lets several annotators code the same session against a shared scheme, and reports stage-level percent agreement and Cohen's kappa between coders.

Distinguishes it from: A general qualitative coding tool. Two commitments separate it: it runs entirely on the researcher's own machine, and it never writes to the source material.

write-up

Definition: A public retrospective account of one solve, posted after a competition. The largest available corpus of solve traces, and the noisiest.

Distinguishes it from: A recorded solve. A write-up is authored after the fact and shows the path the author chose to show, so it evidences that a solve had multiple paths only when the author says so.

playtesting

Definition: Solving one's own challenge, or having someone else solve it, in order to place hints and calibrate difficulty before release.

Distinguishes it from: Piloting the study. Playtesting tunes the artifact; piloting tunes the instrument. Two of the five pilot designers described the first and none described the second.

CTFd

Definition: The competition platform whose submission records supply the anchor events. Named because the anchoring claim depends on the platform actually exporting per-submission timestamps.

Distinguishes it from: The challenge infrastructure, which hosts the artifacts. The platform records attempts; the infrastructure records nothing about the solver.

3 Peripheral

directed hypergraph

Definition: A graph whose edges have sets at both ends, so one edge can consume several questions and emit several. The representation the framework commits to, because an ordinary digraph cannot carry that arity.

Pointer: Gallo et al. 1993 for hyperarcs, hyperpaths and the traversal algorithms, and for the And-Or graph connection, which is the decomposition structure this needs.

hyperpath

Definition: A solution, in the framework's terms. The sub-hypergraph that carries the solver from the opening question to the answered one.

Pointer: The framework's definitions section, where problem, solution and difficulty are defined in that order.

episode

Definition: Schoenfeld's unit for parsing a think-aloud protocol: read, analyze, plan, implement, explore, verify. The direct ancestor of the six phases, and still in use.

Pointer: Schoenfeld 1985, and the recent transfer of the same parsing to language model reasoning traces, which is incidental evidence that the vocabulary survives contact with agents.

chunk

Definition: A domain-specific perceptual unit. The reason expertise does not show up as general ability: masters' recall superiority on chess positions collapses on random boards.

Pointer: Chase and Simon 1973, and the instructional-science twin of the same point, that domain-general problem-solving skills resist teaching.

cognitive load

Definition: The tripartite decomposition of demand into intrinsic, extraneous and germane. The nearest established source for a multidimensional difficulty account.

Pointer: The difficulty vector is a domain adaptation of this and of item response theory's separation of item difficulty from person ability, and is citable as an adaptation rather than as an invention.

option

Definition: A temporally extended action in hierarchical reinforcement learning, discovered rather than named in advance.

Pointer: The open problem on the cluster page, where operations are treated as learned actions instead of a fixed taxonomy. If they can be discovered, the grain problem softens, because the agent finds the grain that pays.

4 Where two locators disagree

Four entries above cite two files that use one term differently, and difficulty cites three. They sit in the entries rather than in a list of errata, because each is a part of the concept that conflicts with another part.

The disagreement	The entry it sits in
The construct is unsettled, and H1 is stated over it	mental operation
Tool selection inside phase three, or above the phases as control	control
Stimulated recall as its own method, or paired with retrospective think-aloud	stimulated recall
Difficulty as a vector, as a relation to what is observable, and as a curve across a set	difficulty

One further gap is worth writing down rather than arguing about. The page’s own index question asks about the era of agent solvers, and the page answers the first half of it and stops. The answer to the second half exists, on the poster and in the designer question bank, and it is the four entries under *The agent in the room* above. Those are the entries that belong on the page.

5 What crosses into the shared lexicon

Terms this project uses that the whole Mental World Modeling cluster uses. Their cluster sense goes in the shared file, and the entries above carry only what Capture-the-Flag fixes about them: mental operation, question, grain, control, phase, elicitation, reliability, difficulty, and typing another agent.

Two of those already differ across the cluster and stay in both files. Grain here means the size of a coded unit of a solve; in the open-problems page it names a problem an agent could solve for itself by discovering its own operations. Difficulty here is a vector over one solver and one problem; in the realism work it is an ordinal priority per dimension of an environment. Neither difference is a defect to be normalised away.

6 Sources

- Every quoted sentence in the Active tier is from the Capture-the-Flag page and the Problem-Solving index of the website, from the SIGCSE Virtual 2026 prep notes, or from the framework document and its section files. The locator on each entry says which.
- Designer statements are from the five pilot interviews of the 2025 autumn round, quoted as reported in `sections/pilot_interviews.tex`.
- The six phases and their contents are from `core idea/sections/part1_operations.tex`, which is the taxonomy of record because it carries the loop amendment and the original writeup does not.

- The three tiers, the three fluency levels and the four fields are from the Tech Fluency page. The reason the second field exists is in `_side/_lexicons/lexicon_method.tex`.
- The framework has not touched data. Every empirical claim above traces to the interviews or to the annotations, and the entries say which.